

SAMATOV DENIS

Machine Learning Engineer | Applied AI Technical Lead

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PROFESSIONAL PROFILE

Machine Learning Engineer and Applied AI Technical Lead with 5+ years of experience building RAG and document-intelligence platforms, medical imaging systems, and scientific ML tools. Leads teams of 3-5 engineers and owns solution architecture, technical delivery, code and solution review, and ML engineering quality. Combines research expertise in tensor methods, Bayesian inference, segmentation, and radiomics with production-oriented Python, FastAPI, testing, observability, and CI/CD practices.

TECHNICAL EXPERTISE

Programming: Python, Git

LLM / RAG: Source-grounded generation, hybrid retrieval, BM25, vector search, embeddings, document parsing and chunking, OCR, provenance, GraphRAG, reranking, LLM evaluation

ML / Computer Vision: PyTorch, scikit-learn, U-Net, Attention U-Net, semantic segmentation, medical imaging, radiomics

Scientific ML: Tucker/HOSVD, tensor decomposition, reduced-order modeling, QR-based sensor placement, ADMM, compressive sensing, Bayesian inference, MCMC

Backend / Infrastructure: FastAPI, Docker, Redis, PostgreSQL, asynchronous workers, observability, automated testing, static analysis, CI/CD

PROFESSIONAL EXPERIENCE

Analytics and Machine Learning Department, MSUU | Machine Learning Engineer / Technical Lead | Apr 2024 - Present

- Lead architecture and delivery of three applied AI platforms - CourseLLM, Modular RAG Platform, and LLM Adviser - coordinating teams of 3-5 engineers and owning technical strategy, system boundaries, reviews, and operational readiness.
- Serve as Technical Lead for LLM Adviser, a five-developer agentic engineering knowledge platform; designed deterministic document processing, traceable outputs, asynchronous execution, and clear boundaries between domain logic and infrastructure.
- Designed reusable ingestion and retrieval components supporting OCR, provenance preservation, hybrid retrieval, parent-document recovery, GraphRAG, reranking, and source-grounded generation.
- Built the supporting execution layer with FastAPI, asynchronous workers, Redis, multi-provider LLM routing, automated evaluation, observability, security controls, and CI quality gates.
- Established repeatable ML/AI engineering practices through mentoring, code reviews, solution reviews, automated testing, and static analysis.

Heriot-Watt TPU Center | Data Scientist / Research ML Engineer | Sep 2024 - Present

- Develop tensor-based reduced-order models for high-dimensional spatiotemporal reservoir data, supporting forecasting and full-field reconstruction from sparse measurements.
- Design QR-based sparse-sensor placement and reconstruction workflows for incomplete observations, including geometry-aware and compressive-sensing variants.
- Published NAB as a reusable Python package with API/CLI workflows, automated tests, reproducible benchmarks, Bayesian parameter-space analysis, convergence diagnostics, and scientific documentation.

Cardiology Research Institute | Machine Learning Engineer | Mar 2021 - May 2025

- Developed an end-to-end CT/MRI segmentation and radiomics application with automated segmentation and manual ROI correction, reducing processing time by 40%.
- Trained U-Net and Attention U-Net models for medical image segmentation and applied radiomic texture analysis to support pathology detection and clinician review.
- Led development of EPIFAT cardiac CT analysis software, integrating anonymized preprocessing, segmentation, radiomics, manual ROI correction, validation, artifact tracking, and cross-platform delivery.
- Implemented patient-level data separation, corrected metric aggregation, reproducible train/resume/test workflows, and versioned artifact lineage, supporting EPIFAT's registration as software No. 2025610317.

EDUCATION

Tomsk Polytechnic University

- Master's Degree, Applied Mathematics and Computer Science | 2024 - 2026
- Bachelor's Degree, Applied Mathematics and Computer Science | 2020 - 2024
- Diploma of Professional Retraining, Data Science and Machine Learning | 2023 - 2024

Skoltech (Skolkovo Institute of Science and Technology)

- Professional Development Course, Large Language Model-Based Agents | 2025
- Professional Development Course, Generative Models Based on Adversarial Learning | 2024

ADDITIONAL INFORMATION

Selected Publications

- "Tensor-Based Modal Decomposition and Sparse Sensor Placement for the Brugge Field Simulation Model." arXiv preprint, 2026.
- "Approach to Identifying Key Areas for Further Reservoir Study Using Tensor-Based Modal Decomposition." Conference paper, 2025.
- "Automatic Segmentation of Epicardial Fat and Quantitative Evaluation of Radiomic Parameters in Cardiac CT." Proceedings of the XXI International Conference "Perspectives of Fundamental Sciences Development," 2024.
- "Capabilities of Radiomic Analysis of Cardiac MRI Images in Cine Mode for Identifying Post-Infarction Areas." Digital Diagnostics.
- "Beam Parameters Restoration at the NICA Accelerator Complex Based on the Beam Position Monitor Data." Proceedings of START, Joint Institute for Nuclear Research, 2023.

Software Registration

- "EPIFAT - Module for Automatic Segmentation of Epicardial Adipose Tissue on Cardiac CT Images." Certificate of State Registration of Computer Program No. 2025610317, issued January 9, 2025.

Selected Awards, Conferences & International Programs

- Accepted Oral Presentation — "Tensor-Based Modal Decomposition with QR Pivoting for Sparse Sensor Placement and Field Reconstruction," AI4X - Accelerate Conference, Singapore, 2026.
- Participant, School-Conference on Tensor Methods in Mathematics and AI, Shenzhen, China, 2024.
- Participant, Skoltech-HIT Summer School, Harbin, China (remote), 2025.
- Accepted Presentation — "Probabilistic Evaluation of Parameter Space Using Neighbourhood Algorithm Bayes," Data Intelligence in the Oil and Gas Industry, Nizhny Novgorod, 2026.
- AI/ML Competition Awards — Winner, FINOdays AI/ML Track; Special Nomination Winner, National Technology Olympiad; Prize Winner, MIPT AI/Math/Physics Hackathon.
- Second-Degree Diploma — Cardiac CT segmentation and radiomics presentation, Perspectives of Fundamental Sciences Development, 2024.